Dhruv Vashist

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Education

Indian Institute of Technology Delhi — New Delhi, India
M.Tech in Computer Science and Engineering; CGPA: 8.9 — *2027 – 2029*

Guru Gobind Singh Indraprastha University — New Delhi, India
B.Tech in Computer Science and Engineering; CGPA: 79.3% — *2021 – 2025*

Experience

Google Summer of Code Contributor — April 2027 – November 2027
Godot Engine — *Remote*

- Fixed a critical rendering bug in C++ and OpenGL that caused certain classes of models to become unusable or to corrupt the model and the surrounding environment entirely.
- Diagnosed the root cause within the engine's rendering pipeline and contributed the fix upstream to the open-source Godot codebase.

Projects

Real-Time Renderer with PBR and Deferred Shading | *C++, Vulkan, VMA, RenderDoc, Nsight Graphics*

- Built a real-time PBR renderer in C++ and Vulkan with deferred shading, cascaded shadow maps, and image-based lighting, rendering 100K+ triangle scenes at 120+ FPS.
- Implemented multi-frame-in-flight synchronization, descriptor indexing, and GPU memory management via VMA, cutting per-frame CPU overhead by 12%.
- Profiled with RenderDoc and Nsight Graphics to identify and fix bandwidth bottlenecks in the G-buffer pass.

GPU-Driven Compute Particle / Fluid Simulation | *C++, Vulkan, GLSL, Dear ImGui*

- Developed a Vulkan compute simulation of 1M particles in real time using spatial hashing, achieving 60 FPS on an RTX-class GPU.
- Used indirect rendering and pipeline barriers to eliminate CPU readback and synchronize compute and graphics queues.
- Integrated Dear ImGui for live parameter tuning and a GPU timestamp-query profiler.

Four Ways – Tactical Multiplayer Game | *C++, TCP Sockets, Client-Server Networking, Anomaly Detection*

- Designed and built the full multiplayer architecture for a 4-player turn-based tactical game inspired by Warhammer tabletop mechanics and Buckshot Roulette-style risk gameplay, using a client-server model over TCP.
- Implemented server-side game state management, turn synchronization, and lobby/matchmaking for 4 concurrent players, keeping state consistent across clients with 15 ms average latency.
- Built an AI-assisted anti-cheat system integrating a pre-trained anomaly detection model to flag impossible moves, abnormal action timing, and tampered game state, combined with server-side validation of all player actions.
- Handled disconnects, reconnection, and desynchronization using a hybrid of state snapshots and rollback.

Technical Skills

Languages: C, C++, HLSL, GLSL
Graphics APIs: Vulkan, OpenGL, DirectX 12
Engines & Tools: Unreal Engine 5, Blender, CMake, RenderDoc, Nsight
Systems & Performance: Parallel Computing, Multithreading, GPU Pipeline, GPU Architecture, Memory and Caching
Other: Reverse Engineering,Computer Network